

DDL, DML and TCL – Easy SQL Notes

SQL commands can be grouped based on what they do in a database. The three important groups here are DDL, DML and TCL.

1. DDL – Data Definition Language

Meaning: DDL is used to create and change the structure of database objects such as tables.

Main DDL Commands

Command	Easy meaning	Example
CREATE	Creates a new table/database object	CREATE TABLE Student (...);
ALTER	Changes the structure of an existing table	ALTER TABLE Student ADD age INT;
DROP	Deletes a table/object completely	DROP TABLE Student;
TRUNCATE	Removes all rows but keeps the table structure	TRUNCATE TABLE Student;

Simple example:

```
CREATE TABLE Student (id INT, name VARCHAR(30));
```

This creates a table named Student with id and name columns.

Remember: DDL mainly deals with the **structure** of the database.

2. DML – Data Manipulation Language

Meaning: DML is used to add, modify, remove and retrieve data stored inside tables.

Command	Easy meaning	Example
INSERT	Adds new data	INSERT INTO Student VALUES (1,'Priya');
UPDATE	Changes existing data	UPDATE Student SET name='Anu' WHERE id=1;
DELETE	Removes selected data	DELETE FROM Student WHERE id=1;
SELECT*	Retrieves/displays data	SELECT * FROM Student;

Simple examples:

```
INSERT INTO Student VALUES (1,'Priya');  
UPDATE Student SET name='Anu' WHERE id=1;  
DELETE FROM Student WHERE id=1;  
SELECT * FROM Student;
```

Remember: DML mainly deals with the **data/records** inside a table.

3. TCL – Transaction Control Language

Meaning: TCL is used to control transactions in a database. A transaction is a group of SQL operations treated as one unit.

Command	Easy meaning
COMMIT	Permanently saves the changes made in a transaction.
ROLLBACK	Cancels changes made after the last COMMIT.
SAVEPOINT	Creates a point to which we can roll back later.

Simple example:

```
UPDATE Student SET name='Priya' WHERE id=1;  
SAVEPOINT s1;  
UPDATE Student SET name='Anu' WHERE id=1;  
ROLLBACK TO s1;  
COMMIT;
```

Here, the second update can be undone using the SAVEPOINT, while the earlier change can be saved with COMMIT.

4. Quick Difference

Type	Works mainly on	Important commands	Easy memory
DDL	Database structure	CREATE, ALTER, DROP, TRUNCATE	Define structure
DML	Data/records	INSERT, UPDATE, DELETE, SELECT	Manage data
TCL	Transactions/changes	COMMIT, ROLLBACK, SAVEPOINT	Control changes

Easy shortcut: DDL = Design/structure, DML = Data, TCL = Transaction control.

Conclusion: DDL builds the database structure, DML works with the data stored in it, and TCL controls whether database changes are saved or undone.